2.2.1 Rectangular form

Definition 2.1 (Rectangular form of a complex number)

A complex number is of the form $x+iy$ **(or** $x+yi$**)**, where x and y are real numbers. x is called the real part and y is called the imaginary part of the complex number.

If $x=0$, the complex number is said to be purely imaginary. If $y=0$, the complex number is said to be real. Zero is the only number which is at once real and purely imaginary. It is customary to denote the standard rectangular form of a complex number $x+iy$ as z and we write $x=\mathrm{Re}(z)$ and $y=\mathrm{Im}(z)$. For instance, $\mathrm{Re}(5-i7)=5$ and $\mathrm{Im}(5-i7)=-7$.

The numbers of the form $\alpha+i\beta,\ \beta\neq 0$ are called imaginary (non real complex) numbers. The equality of complex numbers is defined as follows.

Definition 2.2

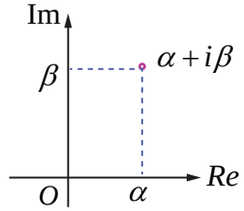
Two complex numbers $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$ are said to be equal if and only if $\mathbf{Re}(z_1)=\mathbf{Re}(z_2)$ **and** $\mathbf{Im}(z_1)=\mathbf{Im}(z_2)$. That is $x_1=x_2$ **and** $y_1=y_2$.

For instance, if $\alpha+i\beta=-7+3i$, then $\alpha=-7$ and $\beta=3$.

2.2.2 Argand plane

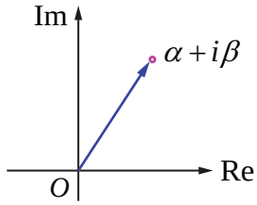
A complex number $z=x+iy$ is uniquely determined by an ordered pair of real numbers (x,y). The numbers $3-8i$, 6 and $-4i$ are equivalent to $(3,-8)$, $(6,0)$, and $(0,-4)$ respectively. In this way we are able to associate a complex number $z=x+iy$ with a point (x,y) in a coordinate plane. If we consider x axis as real axis and y axis as imaginary axis to represent a complex number, then the xy-plane is called complex plane or Argand plane. It is named after the Swiss mathematician Jean Argand (1768 – 1822).

A complex number is represented not only by a point, but also by a position vector pointing from the origin to the point. The number, the point, and the vector will all be denoted by the same letter z. As usual we identify all vectors which can be obtained from each other by parallel displacements. In this chapter, $\mathbb{C}$ denotes the set of all complex numbers. Geometrically, a complex number can be viewed as either a point in $\mathbb{R}^2$ or a vector in the Argand plane.



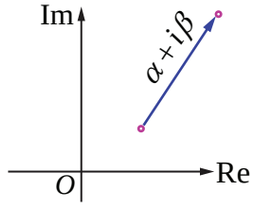
Complex number as a point

Fig. 2.3



Complex number by a position vector pointing from the origin to the point

Fig. 2.4



Complex number as a vector

Fig. 2.5

Illustration 2.1

Here are some complex numbers: $2+i$, $-1+2i$, $3-2i$, $0-2i$, $3+\sqrt{-2}$, $-2-3i$, $\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}$, and $3+0i$. Some of them are plotted in Argand plane.

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